HSI-BERT: Hyperspectral Image Classification Using the Bidirectional Encoder Representation From Transformers

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***Abstract*— Deep learning methods have been widely used in hyperspectral image classification and have achieved state-of-the-art performance. Nonetheless, the existing deep learning methods are restricted by a limited receptive field, inflexibility, and difficult generalization problems in hyperspectral image classification. To solve these problems, we propose HSI-BERT, where BERT stands for bidirectional encoder representations from transformers and HSI stands for hyperspectral imagery. The proposed HSI-BERT has a global receptive field that captures the global dependence among pixels regardless of their spatial distance. HSI-BERT is very flexible and enables the flexible and dynamic input regions. Furthermore, HSI-BERT has good generalization ability because the jointly trained HSI-BERT can be generalized from regions with different shapes without retraining. HSI-BERT is primarily built on a multihead self-attention (MHSA) mechanism in an MHSA layer. Moreover, several attentions are learned by different heads, and each head of the MHSA layer encodes the semantic context-aware representation to obtain discriminative features. Because all head-encoded features are merged, the resulting features exhibit spatial–spectral information that is essential for accurate pixel-level classification. Quantitative and qualitative results demonstrate that HSI-BERT outperforms any other CNN-based model in terms of both classification accuracy and computational time and achieves state-of-the-art performance on three widely used hyperspectral image data sets.**

***Index Terms*— Deep learning, hyperspectral image, image classification, multihead self-attention (MHSA), pattern recognition.**

I. Introduction

HYPERSPECTRAL imagery (HSI) incorporates hundreds of narrow contiguous spectral bands carrying abundant spectral information [1], [2], thus enabling accurate pixel-level classification [3], [4]. The classification tasks of HSI span a broad range of applications, such as ecological science [3], [5], precision agriculture [6]–[8], mineral exploration [9], and medicinel [10]. In the early stage of HSI classification, k-nearest neighbors (k-NNs) [11], support vector machines (SVMs) [12], and the combined pipeline of k-NN and SVM [13] have been introduced for HSI classification. In addition, sparse representation-based methods (SRCs) [14], [15] employed sparse representation algorithms as feature extractors and SVM or k-NN as classifiers. Despite the high spectral resolution, HSIs are originally 3-D and pictorial and have a spatial dependence, which is complementary to spectral behavior [16]. This intrinsic property of HSI requires both spatial and spectral analyses of HSI, which yields a spatial–spectral feature extractor. For example, a Markov random field (MRF)-based feature extractor was introduced to incorporate spatial information with SVM or multinomial logistic regression (MLR) [17], [18], and a generalized composite kernel method for spatial–spectral feature extraction was presented by Li *et al*. [4], which can balance the use of spectral and spatial information without any weighting parameter [2]. In 2014, an edge-preserving filtering-based spectral-spatial classification framework is proposed, which improves the classification accuracy of SVM significantly with much fewer computer resources [19].

Recently, deep learning-based deep spatial–spectral feature extractors [2], [8], [20]–[23] have been widely used by the remote sensing community, and some of them have achieved state-of-the-art performance in HSI classification and deserve special attention. Conventional machine learning techniques require careful engineering and considerable domain expertise to design a feature extractor [24] that transforms the raw HSI data. However, deep learning methods can automatically discover the representations of raw data that are needed for HSI classification. These deep learning models follow an "end-to-end" design philosophy that seeks to avoid explicit structure and "hand-engineering." A general approach to construct deep learning models for remote sensing images was systematically analyzed by Zhu *et al*. [25].

Within these deep learning models, the convolutional neural network (CNN) has drawn intensive attention. There are numerous CNN-based spatial–spectral feature extractors that have been constructed to automatically excavate the HSI

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spatial–spectral information. Slavkovikj *et al*. [26] presented a CNN framework for HSI classification where the spatial–spectral features were extracted from a small neighborhood. Makantasis *et al*. [27] combine a random principal component analysis (RPCA) and CNN into a pipeline (named RPCA-CNN) for spatial–spectral feature extraction. Similar to RPCA-CNN, Yue *et al*. [28] employed the principal component analysis (PCA) followed by CNN to learn the spatial–spectral information. Mei *et al*. [29] introduced a five-layer CNN named SSCNN. Li *et al*. [30] proposed a novel CNN-based method to encode the pixel-pair features and make predictions by a voting strategy. Recently, Lee and Kwon [31] presented deep and wide CNN models (denoted as CDCNN) that can optimally explore contextual interactions by jointly exploiting the local spatial–spectral relationships of the neighboring pixel vectors within a square window. Specifically, the joint exploitation of spatial–spectral information is achieved by a multiscale convolutional filter bank used as an initial component of the proposed CNN pipeline. Furthermore, a novel architecture proposed by Zhang *et al*. [2] exploiting diverse region-based inputs to learn contextual interactional features is shown to have more discriminative power and has achieved superior performance to that of the aforementioned CNN-based methods.

Although CNN-based models have established themselves as powerful methods in HSI classification, they still have three problems. First, the CNN-based feature extractors considerably rely on the convolutional kernels of each layer. When CNN works in the deep layers, the convolutional kernels are shown to extract more high-level features. However, a large kernel size prevents the CNN depth exploitation, so the kernel sizes are generally small (3×3, 5×5); therefore, the small kernel size limits the CNN receptive field.

The CNN cannot combine all the information of a rectangular region in a single kernel. Second, the CNN-based methods are not flexible; they cannot address the arbitrary shape regions. Moreover, since HSI classification is a pixel-level task, the surrounding rectangular region of a target pixel is not the only possible context. In fact, the context with more discriminative power remains to be found. Here, the target pixel means the pixel for which we want to predict a label. Furthermore, since CNN-based methods are not flexible, they cannot be used to find context with more discriminative power and do not enable a dynamic choice of context (different target pixels with different shapes and sizes of context). Third, the CNN-based feature extractors are difficult to generalize. Moreover, the CNN-based models cannot even adapt to the same shape region with different sizes. For example, if a prediction is desired on a 7×7 square region and the CNN is trained on a 9×9 square region, the CNN must not only be retrained on the 7×7 region but also be generalized to different types of regions (namely, generalized from a square region to a round region and a discrete region, and so on).

To address the aforementioned problems of CNN-based methods, we draw inspiration from neural language models. Unlike models in other fields, neural language models are generally flexible and easily generalized [32] due to the needs of processing a vast variety of sentences produced by the combinatorial generalization of words and grammars. Actually, language models can be used in HSI classification with the proper conversion. First, from the spectral aspect, a pixel (spectral vector) in HSI is analogous to a word vector [33] in natural language processing (NLP) because both are distributional representations and both convey some "meaning." The pixel conveys the land cover information, while the word vector conveys the meaning of a word. Second, from the spatial aspect, the context of a target pixel in HSI is analogous to the context of a target word in a corpus. With these two analogies in mind, language models can be used in HSI classification by organizing the input regions to pixel sequences. These flattened pixel sequences (the input regions that have been flattened into pixel sequences) can be directly fed into language models. Motivated by these analogies, we proposed a neural language model-based HSI classification framework called HSI-BERT, where BERT stands for bidirectional encoder representations from transformers [34]; moreover, BERT was originally developed as a language model that achieved the current state-of-the-art performance in 11 language tasks, including sentiment analysis. Thus, HSI-BERT is a method that adapts BERT to HSI feature extraction and classification. HSI-BERT has a global receptive field, so it can capture richer global context information. HSI-BERT achieves this goal by multihead self-attention (MHSA). Given an input region, each head of the MHSA mechanism captures different relationships between any two pixels regardless of their spatial distance. HSI-BERT is much more flexible and can be easily generalized than the CNN-based models. Namely, an arbitrary input region can be processed through our framework (round region, square region, single pixel, and so on), and this property also enables the dynamic choice of context (different target pixels with different shapes and sizes of context). HSI-BERT can not only process different regions but can also be generalized from different regions. For instance, an HSI-BERT jointly trained in some regions can be used to predict regions with different shapes without a significant loss of accuracy. Because of the good generalization ability, HSI-BERT can be used as the pretrained model, and one can fine-tune or directly use HSI-BERT pretrained in some region to make predictions on regions with different shapes. The use of a pretrained model will prevent time costs and computational resources that are associated with retraining the model.

In this article, we proposed a BERT-based hyperspectral image classification framework. To the best of our knowledge, HSI-BERT is the first such framework to adopt BERT in HSI feature extraction and classification. The main contributions of this article are as follows: 1) the global receptive field can be achieved through our framework; 2) a flexible and dynamical choice of the input region is enabled through our framework; and 3) a novel joint training approach is introduced to help HSI-BERT achieve more generalization ability.

II. HSI-BERT

The architecture of HSI-BERT is shown in Fig. 1. Given a target pixel (the pixel for which we want to predict the label), a region that contains the target pixel (context)

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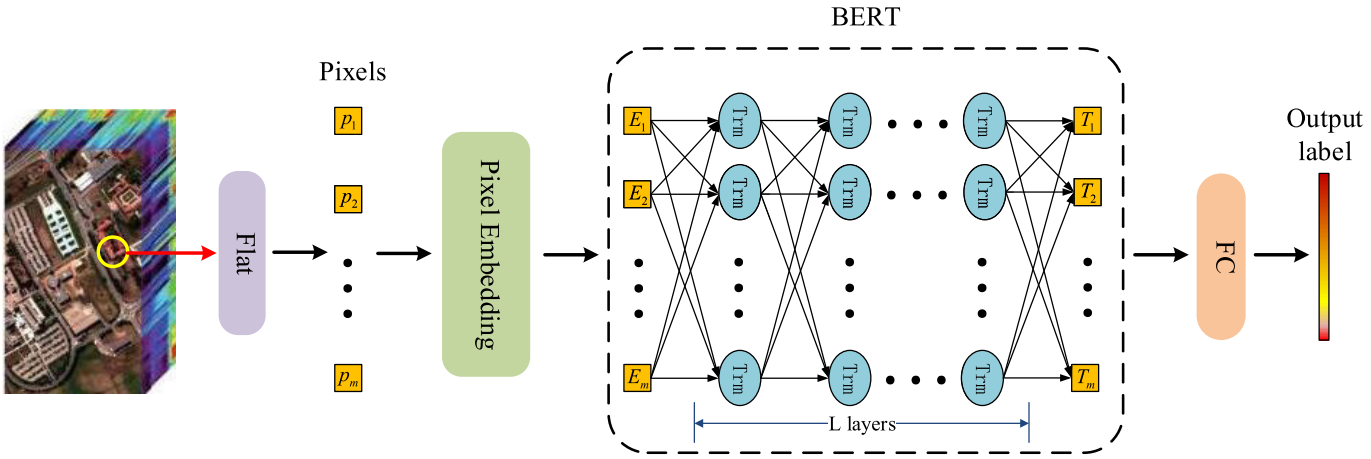


Fig. 1. Proposed architecture. $E_1, E_2, \ldots, E_m$ are the embedded pixels after the pixel embedding (PE) process. Trm is the abbreviation of transformer encoder, while $T_1, T_2, \ldots, T_m$ are the encoded pixel sequence processed through multiple Trm layers. FC is a fully connected layer.

is selected and flattened into a pixel sequence. The pixel sequence is then transformed using the PE block, which transforms pixel vectors to a new vector space with a predefined dimension. The transformed pixel sequence is then directly fed into the BERT module, and the BERT module is a multilayer transformer encoder (Trm) (bidirectional transformer). In HSI-BERT, each pixel will attend to every pixel in a pixel sequence through the MHSA mechanism. The MHSA mechanism learns multiple relations among the pixels in a pixel sequence, with each head learning different relations separately and in parallel. Finally, the BERT learned features are fed into a single-layer fully connected neural network for label prediction.

A. Flat

The Flat operation is applied to organize the selected region into pixel sequence to meet the input requirement of the BERT module. In addition, unless otherwise stated in this article, Flat operation organize the selected regions in raster-scan order, similar to the image transformer [35], PixelRNN [36], and PixelCNN [37].

B. Pixel Embedding

PE aims to transform pixels (spectral vectors) into a new vector space with a predefined dimension (usually smaller dimension). There are two reasons why PE is added to HSI-BERT. The first reason is that PE allows BERT with the desired architecture can be used in all input dimension. Actually, the number of head must be a factor of the input dimension. BERT with the desired architecture cannot be applied to some input dimension without first transforming the input dimension into a suitable dimension. For instance, BERT with the number of head 2 cannot be applied to the University of Pavia data set (with input dimension being 103) without first transforming the input dimension into a suitable dimension. The second reason is that PE can fasten the training speed of BERT by reducing the input dimension (see Fig. 2).

PE consists of an LLT and a learned positional embedding [34], [38], [39]. The LLT works independently and identically on each pixel

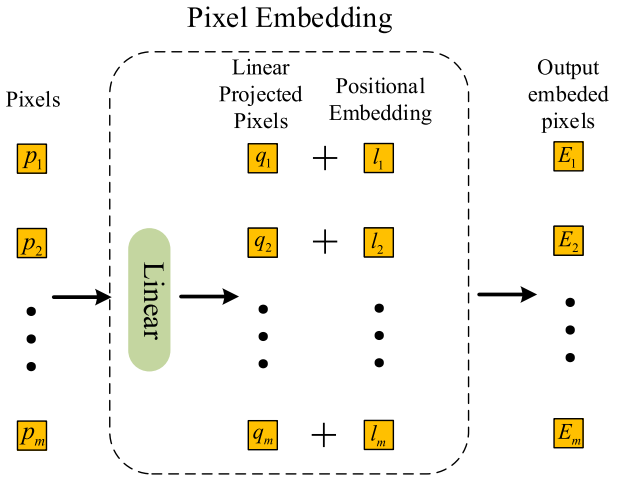


Fig. 2. PE used in this article. The linear projected pixels are pixels processed through the learned linear transformation (LLT), while positional embedding is the learned positional matrix.

$$\mathrm{LLT}(X) = \mathbf{W}^{\top} X \tag{1}$$

where LLT is the abbreviation of learned linear transformation and W is the learned weight of the linear transformation.

Positional embedding is a technique in NLP that is used to encode the positional information of each word in the sentence. Here, the learned positional embedding is used to encode the positional information of the pixel sequence

$$\mathrm{LPE}(X) = X + P \tag{2}$$

where LPE is the abbreviation of learned positional embedding and P is the learned positional matrix (has the same shape with X).

C. BERT

BERT's architecture is a multilayer Trm [34]. In the literature, the Trm is often referred to as a bidirectional transformer. The architecture of a single-layer Trm is shown in Fig. 3.

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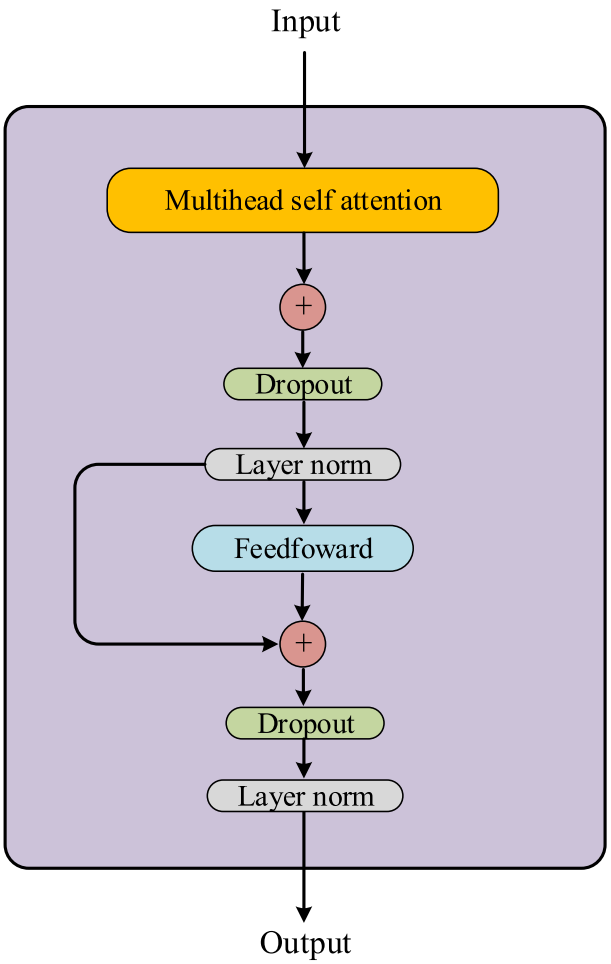


Fig. 3. Trm used in this article.

The self-attentive feedforward sequence model was first introduced in 2017 and achieved state-of-the-art performance in machine translation [38], [39], and it was recently shown to achieve excellent results in other fields, such as image generation [35], question answering [34], [39], and language understanding [34].

1) Multihead Self-Attention: The architecture of the MHSA mechanism of HSI is shown in Fig. 4. An attention function can be described as mapping a query and a set of key-value pairs to an output, where the query, keys, values, and output are all vectors. The output is computed as a weighted sum of the values, where the weight assigned to each value is computed by a compatibility function of the query with the corresponding key [39].

Our attention mechanism is the widely used scaled dot-product attention [38]

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \tag{3}$$

where Q, K, and V represent the query, key, and value, respectively, and d is the dimension of the input data. Our MHSA uses p head $(h_1, h_2, \ldots, h_p)$ and is written as

$$\text{MHSA}(X) = \text{Concat}(h_1, h_2, \ldots, h_p)W^O \tag{4}$$

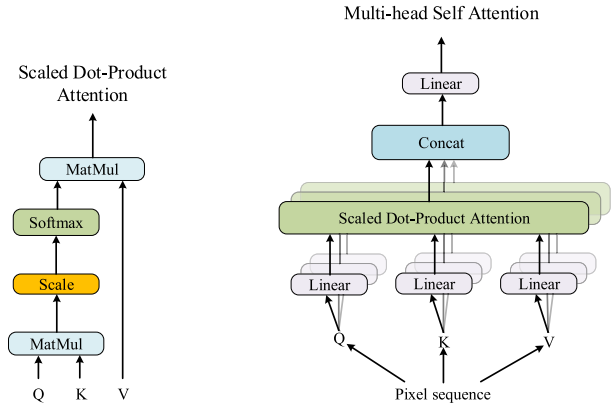


Fig. 4. (Left) Scaled dot-product attention and (Right) MHSA mechanism for the pixel sequence.

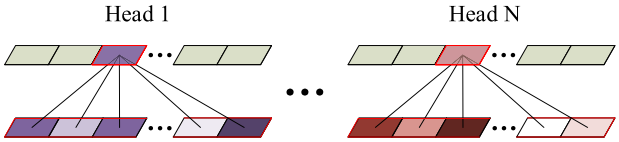


Fig. 5. Visualization of the MHSA mechanism in a flattened arbitrary region. For each head, there is a learned attention distribution over the region, and the deeper the color, the larger the attention value. Different colors represent different attentions.

where W^O represents the learned parameter matrices and

$$h_i = \text{Attention}\left(XW_i^Q, XW_i^K, XW_i^V\right) \tag{5}$$

with affine projections using learned parameter matrices $W^Q \in \mathbb{R}^{d\times d/p}$, $W^K \in \mathbb{R}^{d\times d/p}$, and $W^V \in \mathbb{R}^{d\times d/p}$

Different heads of the MHSA mechanism learn different attentions. All heads work independently and run in parallel. Fig. 5 demonstrated how each head of the MHSA mechanism works in an arbitrary region (which has been flattened into pixel sequence). We have N attention heads in total, and all heads have a global receptive field in the region. All the attention distributions are computed through the scaled dot-product attention shown in Fig. 4.

2) Feedforward: The feedforward module is applied to further transform all heads' learned features through fully connected layers. The feedforward module consists of two linear transformations with a Rectified Linear Unit (ReLU) activation in between. The feedforward module is a shared module across all Trms (the feedforward parameter across different Trms is the same)

$$\text{FFN}(x) = \max(W_1x + b_1, 0)W_2 + b_2. \tag{6}$$

3) Layer Normalization: A normalization method is introduced to reduce the internal covariate shift during the training of deep learning models [40]. The general framework of normalization is given in the following equation:

$$\hat{x} = g \cdot \frac{x - E(x)}{\sqrt{\text{Var}(x) + \varepsilon}} + b \tag{7}$$

where g is the learned scale parameter and b is the learned shift parameter. It is easy to conclude that $E(\hat{x}) = b$ and $\text{Var}(\hat{x}) = g^2$. For layer normalization [41], $x = (x_1, x_2, \ldots, x_d)$ is the input of the current layer, with x_i input

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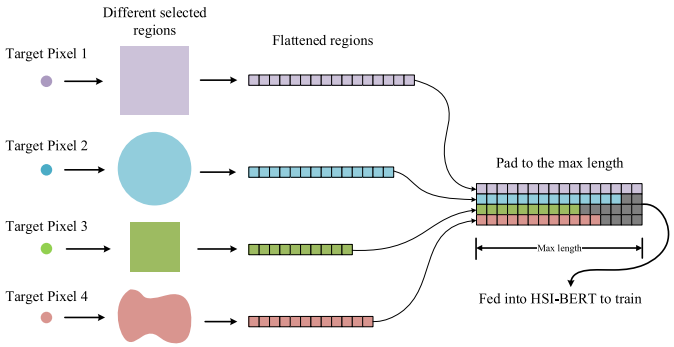


Fig. 6. Data of joint training. The "gray" pixels are dummy pixels used to pad pixel sequences to a max length. Pixel sequences from different regions are concatenated and directly fed into HSI-BERT to train.

to the ith neuron. There are numerous advantages of layer normalization. For instance, it makes the training process much faster, enables a larger learning rate, and serves as a regularizer during training. Unlike batch normalization, it does not require the batch statistics, so it can be used in small batch scenarios.

D. Joint Training of HSI-BERT

HSI-BERT can take arbitrary regions as input, which makes the training very flexible. HSI-BERT can be trained in one single region (single training), just as in CNN-based methods or jointly trained in several different regions. It is worth to note that joint training does not train several different HSI-BERTs simultaneously in different regions but rather trains one HSI-BERT in multiple regions. The aim of joint training is to allow HSI-BERT to be exposed to many different contexts (regions) of the target pixel so that it can be expected to generalize to unseen regions. In order to make a balance between the generalization ability and the training efficiency, each target pixel will be assigned with one region at a training iteration. (Assuming we have three regions {round, square, line} in joint training, we do not select all three regions for each target pixel, rather we simple randomly assign one region for each target pixel, for instance, pixel A assigned with round region, pixel B assigned with square region, and pixel C assigned with line region.)

To train multiple different regions simultaneously in an HSI-BERT, these regions are first flattened into pixel sequence and then padded to a max length with dummy pixels. In this article, we use zero vectors as dummy pixels. This process is shown in Fig. 6. The "pad-to-max length" technique is inspired by language models that pad sentences with a specific token to a max length [42]. After this process, the data can be directly fed into HSI-BERT to train.

E. Analysis on Proposed HSI-BERT

The proposed method introduced in this article is based on BERT. In NLP, BERT can incorporate context from both directions in a sentence, where each token can attend to every token in the self-attention layers of the transformer (global receptive field). Drawing inspirations from NLP, we designed

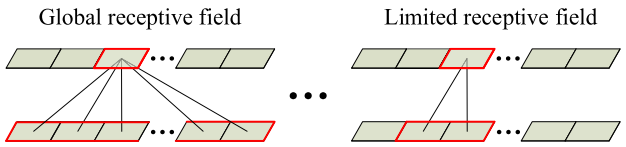


Fig. 7. (Left) Global receptive field versus (Right) limited receptive field in a pixel sequence.

an architecture called HSI-BERT, which uses BERT to learn the feature representation of an arbitrary input region with a global receptive field.

The difference between a global receptive field and a limited receptive field is shown in Fig. 7. A global receptive field captures the global context, while the limited receptive field can only capture the limited context. BERT with a global receptive field captures the global dependence between any two pixels in an input region regardless of how far apart these two pixels are. HSI-BERT achieved this global receptive field with the self-attention layer. For each pixel in an input region, the self-attention layer will compute an attention distribution over the region, and the attention distribution informs the pixel where it should attend to.

The feature extractor of HSI-BERT is a pixel-level feature extractor, which makes it very flexible. The flexibility of HSI-BERT allows arbitrary shapes of an input region, while CNN-based methods only allow a square input region. The flexibility of HSI-BERT also affects the training process. HSI-BERT has two modes of training: one is single training, which trains HSI-BERT in a single region, similar to CNN-based methods and most of the machine learning algorithms, and the other is joint training, which trains an HSI-BERT in multiple different regions. Note that the joint training is not training multiple HSI-BERTs in multiple regions but rather training one HSI-BERT in multiple regions. Our experiment shows that the jointly trained HSI-BERT has good generalization ability, and it can even be generalized to regions with different shapes (namely, from 11×11 and 9×9 square regions generalized to a 7×7 square region and even a round region). Although the joint training approach trains HSI-BERT in multiple regions, joint training does not require more training steps than single training; therefore, joint training does not consume more computational resources than single training.

The good generalization ability of the jointly trained HSI-BERT enables its use as a pretrained model for HSI feature extraction and classification. Namely, HSI-BERT pretrained in some region can be fine-tuned or directly used to make predictions on a region with a different shape. The use of the pretrained model will prevent the time costs and computational resources associated with retraining the model.

III. Experiments and Analysis

The proposed HSI-BERT is evaluated in five different experiments. The first experiment aims to evaluate how the model depth and the number of attention heads affect HSI-BERT. The second experiment aims to evaluate the flexibility of HSI-BERT, which enables the flexible shape of input regions.

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TABLE I
DATA SPLITTING FOR TRAINING AND TESTING OF THE INDIAN PINES DATA SET

No.	Class	Train	Test
1	Corn-notill	200	1228
2	Corn-mintill	200	630
3	Grass-pasture	200	283
4	Hay-windrowed	200	278
5	Soybean-notill	200	772
6	Soybean-mintill	200	2255
7	Soybean-clean	200	393
8	Woods	200	1065
	TOTAL	1600	6904

In this experiment, multiple different HSI-BERTs are trained in regions with different shapes and then evaluated in the test set. The third experiment, in contrast, trains one HSI-BERT in multiple regions with different shapes (joint training) and aims to evaluate the generalization ability of the jointly trained HSI-BERT. The fourth experiment, which is an ablation experiment on the PE module, aims to validate the functionality of PE. In the fifth and last experiment, the classification performance and computational time of the proposed HSI-BERT are compared with several state-of-the-art HSI classification approaches. For edge pixels (the boundary of HSI), we pad them with zeros to a target shape (square, round, and so on).

For better estimation of parameters and classification performance, the same data augmentation strategy introduced in DRCNN [2] is applied, which augments the training data by a factor of two through flipping and adds small Gaussian noise to the original HSI data.

A. Experimental Data Sets

The proposed HSI-BERT is evaluated on three widely used hyperspectral image data sets; they are the Indian Pines, University of Pavia, and Salinas data sets, as illustrated in Fig. 8. For each data set, the training data composed by randomly sampled 200 labeled pixels per class from the ground truth map, while the rest are preserved for testing.

The Indian Pines data set contains 145×145 pixels gathered by the AVIRIS sensor in Northwestern Indiana, where AVRIS stands for airborne visible infrared imaging spectrometer. The original Indian Pines data set contains 220 spectral channels in the wavelength range from 0.4 to 2.5 μm with a spatial resolution of 20 μm. In this article, 20 bands corrupted by water absorption effects are discarded. Some small classes are also discarded from the original 16 classes [12], [43]. The numbers of training and testing samples are listed in Table I.

The University of Pavia data set contains 610×340 pixels collected by the ROSIS sensor at the University of Pavia, where ROSIS stands for reflective optics system imaging spectrometer. This image scene contains 103 spectral bands in the wavelength range from 0.43 to 0.86 μm with a spatial resolution of 1.3 m. The data set was provided by Prof. Paolo Gamba from the Telecommunications and Remote Sensing Laboratory, University of Pavia. It contains nine classes and 42 776 labeled pixels in total, and the numbers of training and testing samples are listed in Table II.

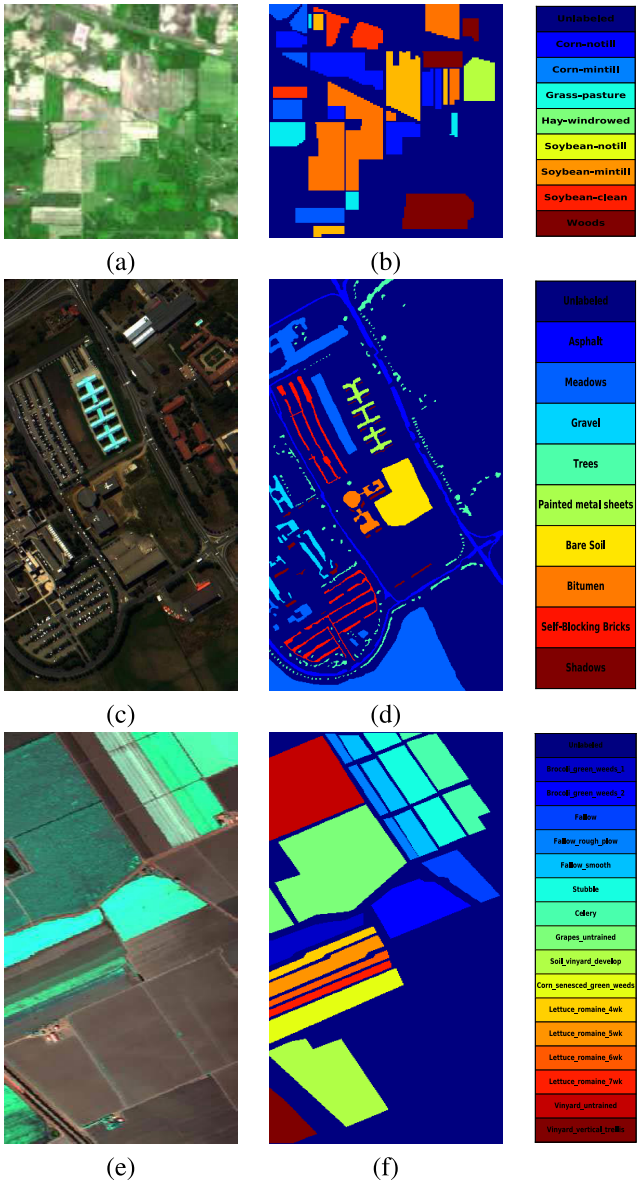


Fig. 8. Three hyperspectral image data sets. (a) False-color Indian Pines image. (b) Ground-truth map of the Indian Pines data set. (c) False-color University of Pavia image. (d) Ground-truth map of the University of Pavia data set. (e) False-color Salinas image. (f) Ground-truth map of the Salinas data set.

The Salinas data set contains 512×217 pixels also collected by the AVIRIS sensor over Salinas Valley, California. These data contain 224 spectral bands with a spatial resolution of 3.7 m. There are 16 classes in total, and the number of training and testing samples are listed in Table III.

B. Analysis on Model Depth and Attention Heads

The model depth is the number of Trm layers in HSI-BERT. Both the model depth and the number of attention heads control the model complexity of HSI-BERT. High complexity may cause overfitting and the waste of computational resources;

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TABLE II
DATA SPLITTING FOR TRAINING AND TESTING OF THE PAVIA UNIVERSITY DATA SET

No.	Class	Train	Test
1	Asphalt	200	6431
2	Meadows	200	18449
3	Gravel	200	1899
4	Trees	200	2864
5	Painted metal sheets	200	1145
6	Bare Soil	200	4829
7	Bitumen	200	1130
8	Self-Blocking Bricks	200	3482
9	Shadows	200	747
	TOTAL	1800	40976

TABLE III
DATA SPLITTING FOR TRAINING AND TESTING OF THE SALINAS DATA SET

No.	Class	Train	Test
1	Broccoli green weeds 1	200	1909
2	Broccoli green weeds 2	200	3526
3	Fallow	200	1776
4	Fallow rough plow	200	1194
5	Fallow smooth	200	2478
6	Stubble	200	3759
7	Celery	200	3379
8	Grapes untrained	200	11071
9	Soil vineyard develop	200	6003
10	Corn senesced green weeds	200	3078
11	Lettuce romaine 4 wk	200	868
12	Lettuce romaine 5 wk	200	1727
13	Lettuce romaine 6 wk	200	716
14	Lettuce romaine 7 wk	200	870
15	Vineyard untrained	200	7068
16	Vineyard vertical trellis	200	1607
	TOTAL	3200	50929

TABLE IV
CLASSIFICATION RESULTS OF HSI-BERT IN DIFFERENT MODEL DEPTHS

Number Layers	Indian Pines	Pavia University	Salinas
1	98.93	**99.75**	98.91
2	**99.56**	99.66	**99.40**
3	99.46	99.57	99.23

however, low complexity causes underfitting. Our goal is to find the smallest model depth and number of attention heads without causing underfitting. First, three HSI-BERTs are trained by fixing all other parameters and vary the number of model depths from 1 to 3. The overall accuracy in three data sets is listed in Table IV. The best model depths for the Indian Pines, Pavia University, and Salinas data sets are 2, 1, and 2, respectively. It is obvious that when the model depth is 1, HSI-BERT underfits the Indian Pines and Salinas data sets. Then, HSI-BERTs are trained by fixing all other parameters and varying the number of heads, and the classification accuracy is listed in Table V. The best number of attention heads for the Indian Pines, Pavia University, and Salinas data sets is 2, 4, and 1, respectively.

TABLE V
CLASSIFICATION RESULTS OF HSI-BERT IN DIFFERENT NUMBERS OF HEADS

Number Head	Indian Pines	Pavia University	Salinas
1	99.45	99.66	**99.56**
2	**99.56**	99.64	99.40
4	99.49	**99.67**	99.41
10	99.46	99.64	99.34

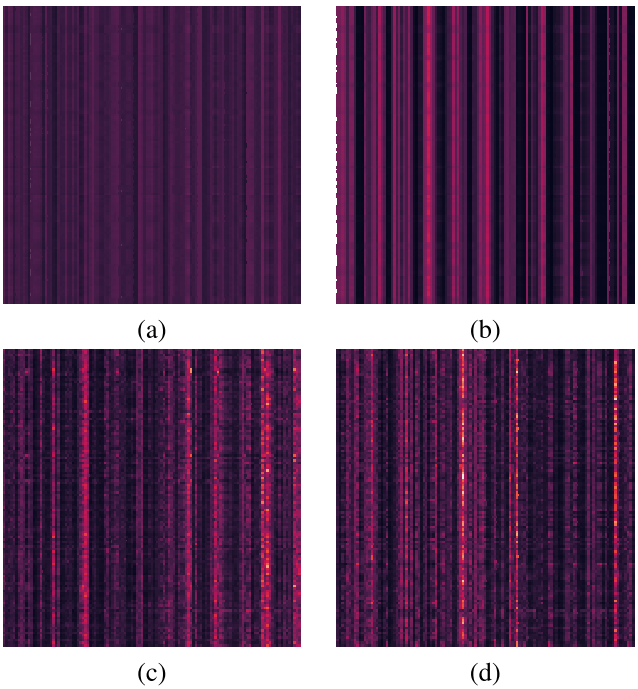


Fig. 9. Self-attention visualization of HSI-BERT. (a) Attention of layer 1, head 1. (b) Attention of layer 1, head 2. (c) Attention of layer 2, head 1. (d) Attention of layer 2, head 2. Each attention matrix has the shape of 121×121, since an 11×11 square has 121 pixels. The ith column of each attention matrix corresponds to the attention distribution over the whole pixel sequence of the ith pixel.

In this experiment, only the 11×11 square region is used to train the HSI-BERT. The visualization of the attention distribution of different layers and heads on a randomly sampled 11×11 square region is shown in Fig. 9. It is obvious that the attention distribution of a pixel (each column of those attention matrices) in layer 1 is less diverse than that of layer 2, which also suggests that HSI-BERT starts to learn more complex attentions as the model depth grows.

C. Evaluation of the Flexibility of HSI-BERT

The "flexibility" means the ability that enables the flexible choice of input regions. HSI-BERT can take arbitrary regions as input, and the regions used in this experiment are shown in Fig. 10. These regions are dot, horizontal line, vertical line, cross, square, and round regions. For each region, one HSI-BERT is trained. The region is first flattened into a pixel

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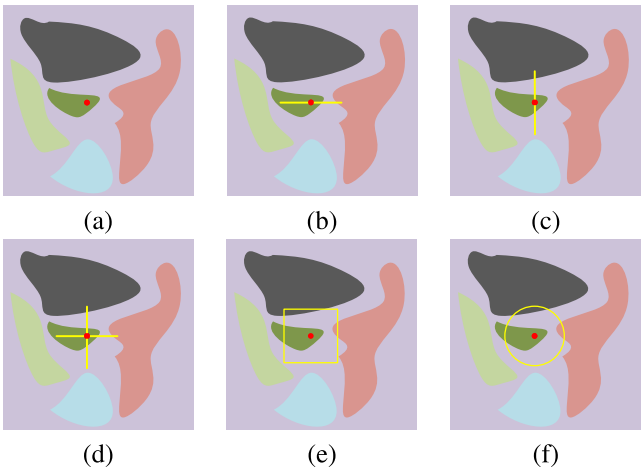


Fig. 10. Several different regions used in the experiment. The red point represents the target pixel, and the region enclosed by the yellow line together with its boundary is the selected region. (a) Dot region, contains only the target pixel. (b) Horizontal line region, with target pixel in the center. (c) Vertical line region, with target pixel in the center. (d) Cross region, with target pixel in the center. (e) Square region, with target pixel in the center. (f) Round region, with target pixel in the center.

TABLE VI
OVERALL ACCURACY OF HSI-BERT IN DIFFERENT REGIONS (WITHOUT DATA AUGMENTATION)

Region type	Indian Pines	Pavia University	Salinas
dot	81.79	90.13	91.05
horizontal line	95.94	98.49	96.53
vertical line	96.42	98.13	96.67
cross	97.71	98.93	96.77
square	98.87	99.74	98.65
round	98.80	99.70	98.60

sequence and then fed into HSI-BERT to train. The only training mode used in this experiment is single training that trains one HSI-BERT in a specific region. Therefore, there is only one region encountered in the training process for each region (the training data were already aligned), and there is no need to further process the training data with an additional "pad-to-max length" technique.

In the experiment, both the horizontal line and vertical line contain 11 pixels, including the target pixel. The cross region composed of a horizontal line, and a vertical line contains 22 pixels in total (for simplification, the duplicated target pixel is not removed). The square region has the shape of 11×11 and thus contains 121 pixels. The round region has a radius of 6 and contains 113 pixels. For all these regions, the same hyperparameters are applied. Namely, the number of head is 2 and the number of layer is 2, while the start learning rate is $3e^{-4}$, the number of training epochs is 200, the batch size is 128, and the dropout rate is 0.2. The classification performances in different regions are shown in Table VI. This experiment suggests that HSI-BERT enables a flexible choice of input regions.

Note that the cross region uses only 18.2% pixels of an 11×11 square region, while it suffers less than 2% accuracy loss, which suggests that the cross region may have relatively higher data efficiency. Since HSI-BERT enables the flexible choice of regions, regions with more data efficiency can be explored in future work.

D. Evaluation of the Generalization Ability of HSI-BERT

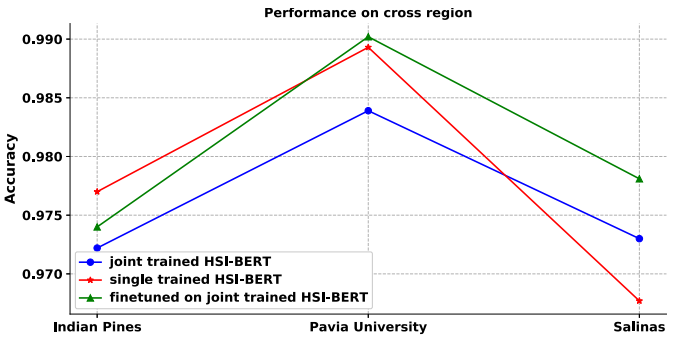
In this experiment, the jointly trained HSI-BERT is evaluated. Again, the aim of joint training is to allow HSI-BERT to be exposed to many different contexts (regions) of the target pixel and expect that it can be generalized to the unseen regions. To validate that, HSI-BERT needs to be trained in some regions and tested in totally different regions. In the experiment, an HSI-BERT is jointed trained in 5×5, 7×7, and 9×9 square regions. Since there are multiple different regions encountered in the joint training process, in order to align the training data, the training data are processed with an additional "pad-to-max length" technique described in Fig. 6. Then, the jointed trained HSI-BERT is tested in four different test regions, including the cross region, round regions with radii 3 and 6, and 11×11 square region. The jointly trained HSI-BERT is also compared with those HSI-BERTs specifically trained (single trained) in the aforementioned test regions. The jointly trained HSI-BERT achieved reasonably good classification performance with only 1% accuracy loss on average when compared with those HSI-BERTs specifically trained in test regions. Finally, the jointly trained HSI-BERT is fine-tuned with an extra ten iterations, and the fine-tuned HSI-BERT achieves nearly the same and sometimes higher classification accuracy than that of the HSI-BERTs specifically trained in test regions.

All these jointly trained or nonjointly trained HSI-BERT share the same hyperparameters. Namely, the number of heads is 2, the number of layers is 2, the start learning rate is $3e^{-4}$, the number of training epochs is 200, the batch size is 128, and the dropout rate is 0.2. The experimental results are shown in Fig. 11.

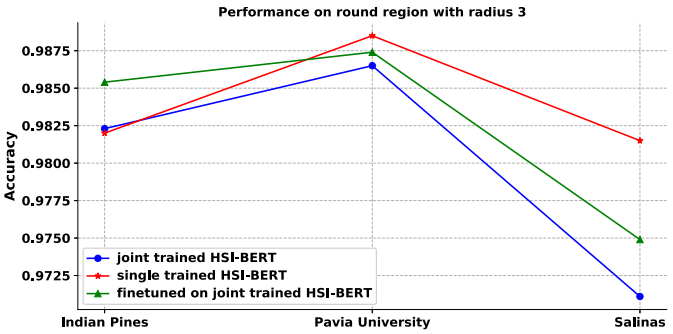
E. Evaluation of Pixel Embedding

This experiment aims to evaluate the functionally of PE. As mentioned earlier, there are two reasons why PE is added to HSI-BERT. The first reason is that PE allows BERT with the desired architecture can be used in all input dimension, and the second reason is that PE can fasten the training speed of BERT by reducing the input dimension (see Fig. 2). As we analyzed in Section III-B, the best architecture of BERT on the University of Pavia data set has two attention heads. However, we cannot apply the best architecture directly on the University of Pavia data set without mapping the data set into a suitable dimension. Without PE, the only possible architecture of BERT has one attention head. With PE, the best architecture of BERT can be applied, which yields 99.73% classification accuracy in the University of Pavia data set. Another notable effect of PE is that it can fasten the training speed because it reduces the input dimension, which makes the computation of attention much cheaper. In this experiment, the PE reduces the dimension of the Indian pines, University of Pavia, and Salinas data set into 100, 50, and 100, respectively (see Table VII).

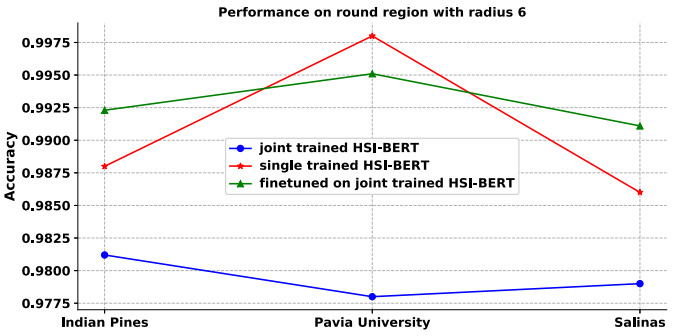
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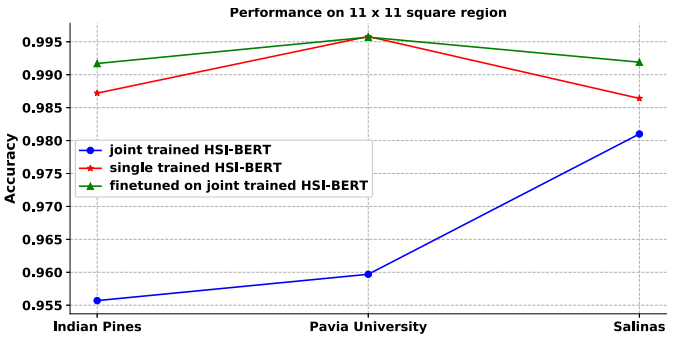
(a)



(b)



(c)



(d)

Fig. 11. Generalization ability of jointly trained HSI-BERT. Performance of jointly trained HSI-BERT on (a) cross region, (b) round region with radius 3, (c) round region with radius 6, and (d) 11 × 11 square region.

F. Comparison With Other Algorithms

This experiment aims to compare the performance of the proposed HSI-BERT with some traditional methods and CNN-based methods. For traditional methods, HSI-BERT is compared with the SVM with radial basis function (SVM-RBF), SVM-RFS [44] (multiple classifier systems-based SVM with random feature selection), and SVM-MRF [12]. For CNN-based methods, HSI-BERT is compared with CNN [45], RPCA-CNN [27], CNN-PPF [30], CDCNN [31], SSCNN [29], and DRCNN [2]. The proposed HSI-BERT was trained and evaluated using ten randomly sampled training and testing data, and the average results and their standard deviations are recorded. For CNN-based methods and the proposed HSI-BERT, the region used for training is an 11 × 11 square.

TABLE VII

COMPARISON OF CLASSIFICATION ACCURACY (%) AND TRAINING TIME (h: HOURS) WITH OR WITHOUT PE

Dataset		OA(%)	Training time(h)
Indian Pines	With PE	99.55	0.117
	Without PE	99.48	0.187
University of Pavia	With PE	99.73	0.110
	Without PE	99.55	0.111
Salinas	With PE	99.45	0.245
	Without PE	99.32	0.513

TABLE VIII

CLASSIFICATION ACCURACY (%) IN DIFFERENT NUMBERS OF TRAINING DATA FOR THE PROPOSED HSI-BERT AND CNN BASELINES

Dataset	Method	50	100	150	200
Indian Pines	CNN[45]	80.43	84.32	85.3	87.01
	CNN-PPF[30]	88.34	91.72	93.14	93.9
	CDCNN[31]	84.43	88.27	-	94.24
	DRCNN[2]	88.74	94.94	97.49	98.54
	HSI-BERT	**91.75**	**97.21**	**98.14**	**99.56**
University of Pavia	CNN[45]	86.39	88.53	90.89	92.27
	CNN-PPF[30]	88.14	93.35	94.97	96.48
	CDCNN[31]	92.19	93.55	-	96.73
	DRCNN[2]	96.91	98.67	99.21	99.56
	HSI-BERT	**98.25**	**98.95**	**99.39**	**99.75**
Salinas	CNN[45]	89.20	89.58	89.60	89.72
	CNN-PPF[30]	92.15	93.88	93.84	94.80
	CDCNN[31]	82.74	98.58	-	95.42
	DRCNN[2]	93.46	95.54	97.36	98.33
	HSI-BERT	**96.30**	**98.56**	**99.23**	**99.56**

TABLE IX

COMPARISON OF TRAINING AND TESTING TIME (h: HOURS AND s: SECONDS) ON THREE HYPERSPECTRAL IMAGE DATA SETS (IP: INDIAN PINES, UP: UNIVERSITY OF PAVIA, AND SA: SALINAS)

Method		IN	UP	SA
CNN[45]	Training(h)	**0.0126**	**0.0124**	**0.0214**
	Testing(s)	**0.45**	**0.79**	**0.62**
CNN-PPF[30]	Training(h)	12.81	2.14	28.65
	Testing(s)	10.16	36.21	50.05
DRCNN[2]	Training(h)	1.58	0.92	3.39
	Testing(s)	83.27	224.65	572.96
HSI-BERT	Training(h)	0.12	0.07	0.21
	Testing(s)	3.52	9.28	21.00

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TABLE X
CLASSIFICATION ACCURACIES (%) IN THE INDIAN PINES DATA SET

	SVM-RBF	SVM-RFS	SVM-MRF	CNN	RPCA-CNN	CNN-PPF	CDCNN	SSCNN	DRCNN	HSI-BERT
OA	82.98	88.68	95.34	87.01	91.13	93.9	94.24	96.63	98.54	$\mathbf{99.56 \pm 0.085}$
AA	88.19	92.55	95.02	90.60	90.49	96.38	95.75	96.29	99.29	$\mathbf{99.72 \pm 0.054}$
1	76.14	88.73	93.55	78.58	82.39	92.99	90.1	96.28	98.20	**98.77**
2	85.40	91.20	90.04	85.24	85.41	96.66	97.1	92.26	99.79	**99.81**
3	97.88	97.52	95.80	96.10	95.24	98.58	100	99.3	100	100
4	99.28	100	100	99.64	100	100	100	100	100	100
5	83.94	91.67	91.12	89.64	82.76	96.24	95.9	92.84	99.78	**99.83**
6	73.48	78.79	97.72	81.55	96.2	87.8	87.1	98.21	96.69	**99.48**
7	92.11	93.76	91.71	95.42	82.14	98.98	96.4	92.45	99.86	**100**
8	97.28	98.74	99.84	98.59	99.81	99.81	99.4	98.98	**99.99**	99.91

Tables IX–XII provide a quantitative classification assessment using the Indian Pines, University of Pavia, and Salinas data sets, respectively. From all results in the three data sets, it is easy to conclude that the proposed HSI-BERT outperforms all the other nine methods, including all CNN-based methods. For example, in Table X, the proposed HSI-BERT yields an accuracy of 99.56% and achieved 1.02% absolute improvement over the best CNN-based method. In summary, the proposed HSI-BERT is able to consistently outperform the best CNN baseline with an average improvement of +0.81 and +0.45 for the OA and AA metrics, respectively. For illustrative purposes, Figs. 12–14 present classification maps of CNN-based methods and HSI-BERT; these classification maps are consistent with the results listed in Tables IX–XII. It is obvious that the classification maps of HSI-BERT are less noisy and achieved less misclassification than all four CNN-based methods. For example, in Fig. 13, the soybean-mintill region is obviously less noisy and achieved more accurate classification than the other methods, and the same situation also exists in Fig. 15 in the grapes untrained region, and so on.

Furthermore, to evaluate the performance of HSI-BERT in a small training data scenario, HSI-BERT is also evaluated in different training samples per class. Table VIII lists the classification performance with different numbers of training data. These training data are composed of randomly sampled 50 samples per class, 100 samples per class, 150 samples per class, and 200 samples per class from the original data, while the remained data are retained as test data. Obviously, for any methods, the smaller the training data, the worse the classification accuracy is. It is easy to see that the proposed method performs consistently better than all four state-of-the-art CNN-based methods. Therefore, another important observation is that HSI-BERT has a special advantage over CNN-based methods in small training data scenarios. Actually, the smaller the training data size, the larger the advantage is. For instance, when the training data size is 100 per class, the proposed HSI-BERT outperforms the best CNN baseline with an average OA improvement of +1.86, but when the training data size changes to 50, the average OA improvement also increases to +2.39.

In addition to the good classification performance, the proposed HSI-BERT is also much faster than some spectral-spatial CNN-based methods in the training phase. The training and testing time of the proposed HSI-BERT, DRCNN [2] CNN [45], and CNN-PPF [30] are summarized in Table IX. In the training phase, CNN [45] is much faster than the other

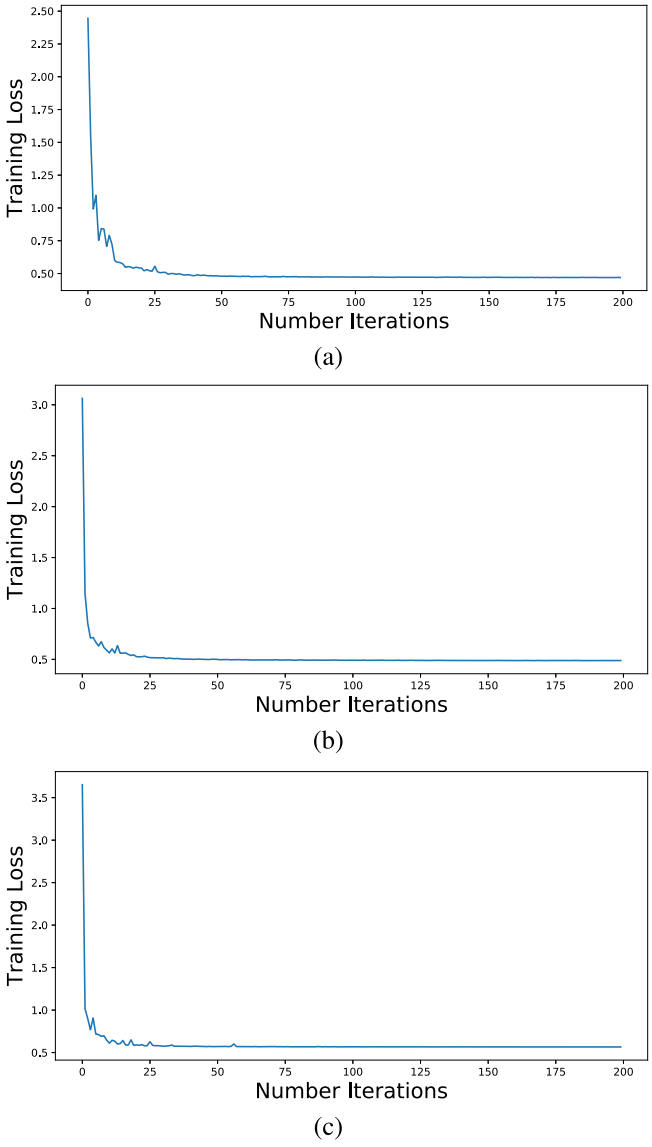


Fig. 12. Training loss curve (convergence curves) of three hyperspectral image data sets. (a) Indian Pines data set, (b) University of Pavia data set, and (c) Salinas data set.

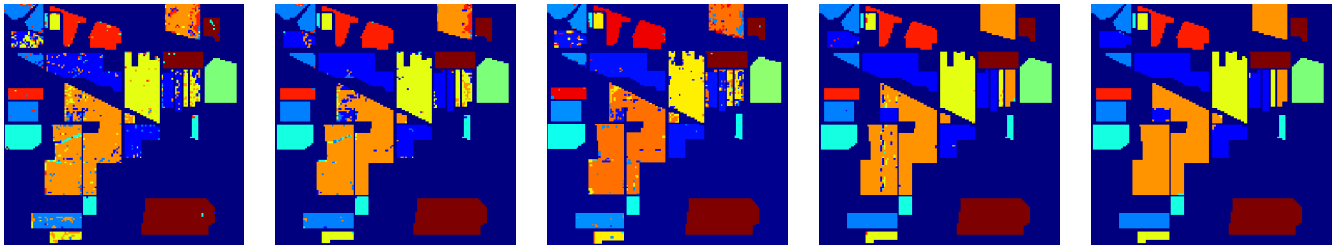
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TABLE XI
CLASSIFICATION ACCURACIES (%) IN THE UNIVERSITY OF PAVIA DATA SET

	SVM-RBF	SVM-RFS	SVM-MRF	CNN	RPCA-CNN	CNN-PPF	CDCNN	SSCNN	DRCNN	HSI-BERT
OA	89.24	91.1	92.63	92.27	93.87	96.48	96.73	98.41	99.56	$\mathbf{99.75 \pm 0.063}$
AA	91.82	92.92	94.04	93.36	94.38	97.03	95.77	98.22	99.53	$\mathbf{99.86 \pm 0.069}$
1	84.01	87.95	98.22	88.38	92.43	97.42	94.6	97.4	98.43	**99.68**
2	88.9	91.17	98.90	91.27	94.84	95.76	96	99.4	99.45	**99.64**
3	87.57	86.99	88.97	85.88	90.89	94.05	95.5	94.84	99.14	**99.82**
4	96.09	95.5	93.64	97.24	93.99	97.52	95.9	99.16	99.50	**99.70**
5	99.91	99.85	99.11	99.91	100	100	100	100	100	100
6	93.33	94.31	80.13	96.41	92.86	99.13	94.1	98.7	**100**	99.98
7	93.98	94.74	82.79	93.62	93.89	96.19	97.5	100	99.70	100
8	82.94	85.89	91.88	87.45	91.18	93.62	88.8	94.57	99.55	**99.94**
9	99.6	99.89	100	99.57	99.33	99.6	99.5	99.87	**100**	99.99

TABLE XII
CLASSIFICATION ACCURACIES (%) IN THE SALINAS DATA SET

	SVM-RBF	SVM-RFS	SVM-MRF	CNN	RPCA-CNN	CNN-PPF	CDCNN	SSCNN	DRCNN	HSI-BERT
OA	81.55	93.15	94.59	89.28	92.39	94.8	95.42	97.42	98.33	$\mathbf{99.56 \pm 0.089}$
AA	89.33	96.87	97.32	94.84	96.83	97.73	98.28	98.94	99.26	$\mathbf{99.84 \pm 0.022}$
1	96.81	99.55	100	97.34	98.84	100	100	100	100	100
2	94.67	99.92	99.70	99.29	99.61	99.88	100	99.89	100	100
3	90.27	99.44	98.94	96.51	99.75	99.60	100	99.89	99.98	**100**
4	98.61	99.86	98.44	99.66	98.79	99.49	99.3	99.25	99.89	**100**
5	94.82	98.02	99.47	96.97	99.84	98.34	98.5	99.39	99.83	**99.92**
6	97.61	99.7	99.95	99.60	99.7	99.97	100	100	100	100
7	99.24	99.69	100	99.49	79.05	100	99.8	99.82	99.96	99.96
8	54.69	84.85	87.64	72.25	99.17	88.68	83.4	91.45	94.14	**98.48**
9	98.32	99.58	99.45	97.53	96.88	98.33	99.6	99.95	99.99	**100**
10	81.91	96.49	94.41	91.29	99.31	98.6	94.6	98.51	99.20	**99.93**
11	90.57	98.78	99.91	97.58	100	99.54	99.3	99.31	99.99	**100**
12	92.43	100	99.64	100	100	100	100	100	100	100
13	98.07	99.13	100	99.02	98.97	99.44	100	99.72	100	100
14	90.39	98.97	98.79	95.05	82.24	98.96	100	100	100	100
15	60.06	76.38	83.37	76.83	97.57	83.53	100	96.24	95.52	**99.26**
16	90.87	99.56	97.34	98.94	99.61	99.31	98	99.63	99.72	**99.97**



(a) (b) (c) (d) (e)

Fig. 13. Visualization of classification maps of the proposed HSI-BERT and CNN methods on the Indian Pines data set. (a) CNN: 87.02%. (b) CNN-PPF: 93.91%. (c) CDCNN: 94.23%. (d) DRCNN: 98.55%. (e) HSI-BERT: 99.56%.

three because CNN [45] is not a spatial–spectral classification algorithm since it does not use spatial information. As for spatial–spectral classification algorithms, HSI-BERT is much faster than DRCNN and CNN-PPF. For instance, HSI-BERT is 106 and 13 times faster than CNN-PPF and DRCNN, respectively, in the Indian Pines data set, and similar situations exist when using the other two data sets. One reason why HSI-BERT is so fast in the training phase is due to the relatively shallow architecture (one layer and two layers), and unlike other deep learning methods, it does not require

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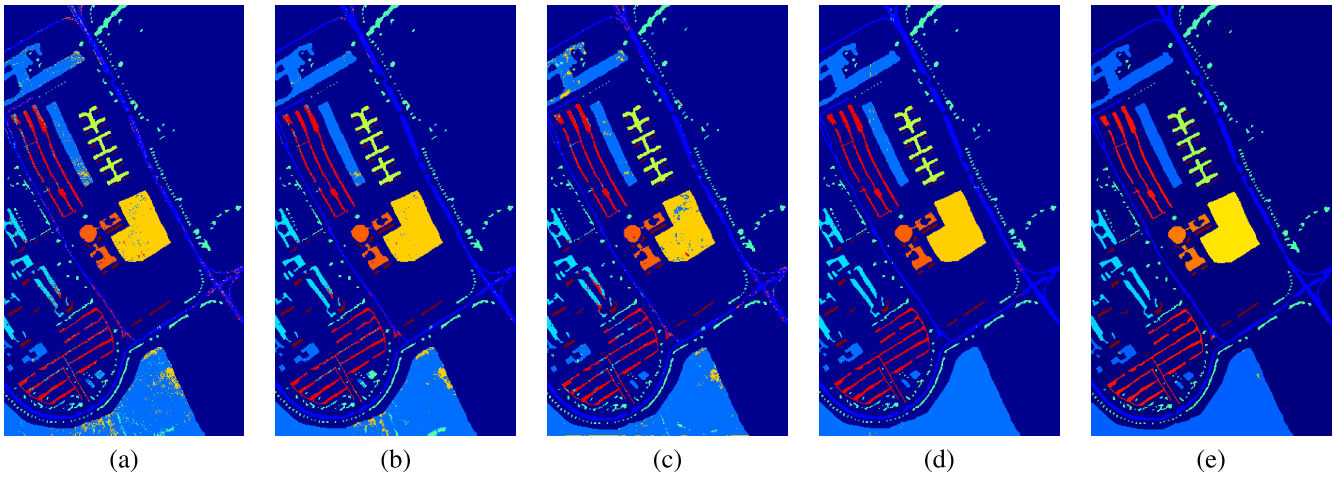


Fig. 14. Visualization of classification maps of the proposed HSI-BERT and CNN methods on the University of Pavia data set. (a) CNN: 92.28%. (b) CNN-PPF: 96.49%. (c) CDCNN: 96.73%. (d) DRCNN: 99.57%. (e) HSI-BERT: 99.75%.

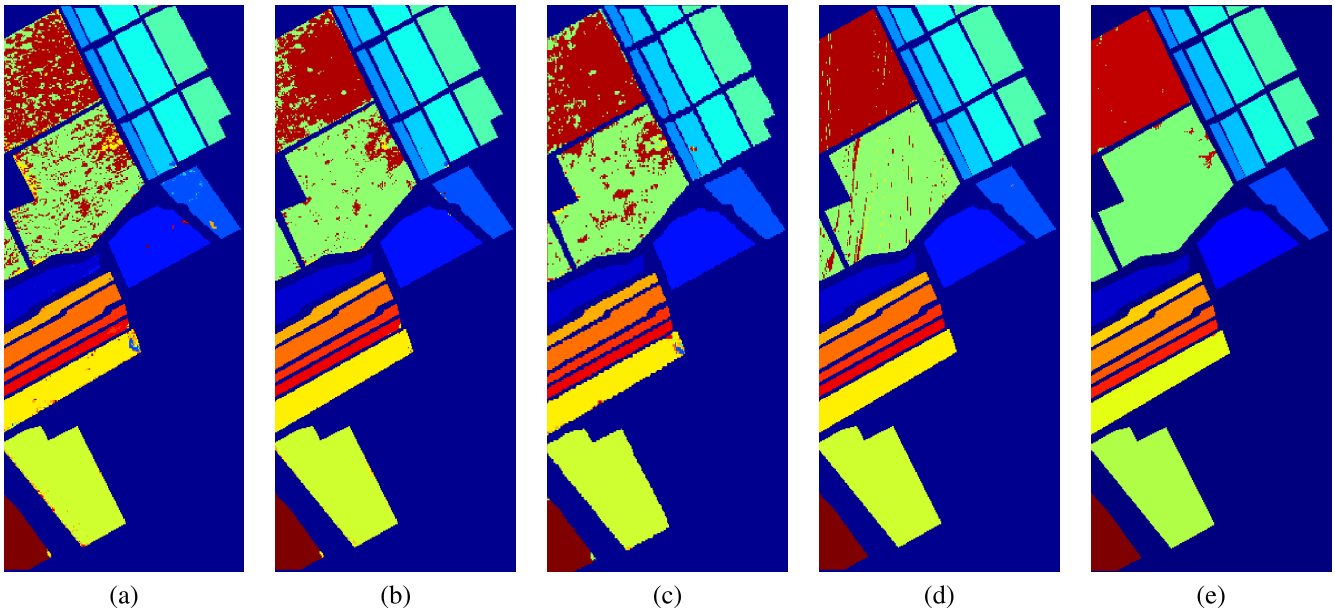


Fig. 15. Visualization of classification maps of the proposed HSI-BERT and CNN methods on the Salinas data set. (a) CNN: 89.29%. (b) CNN-PPF: 94.84%. (c) CDCNN: 95.42%. (d) DRCNN: 98.33%. (e) HSI-BERT: 99.56%.

a very deep architecture to achieve satisfactory performance. Moreover, the other reason is due to the parallel implementation of each head in the MHSA mechanism of HSI-BERT. In the testing phase, the CNN [45] is much faster than the other methods. The proposed HSI-BERT is slower than the CNN [45] but faster than the other baselines. Moreover, from the convergence curves shown in Fig. 12, it is easy to conclude that the HSI-BERT can converge in approximately 100 iterations. The training time of HSI-BERT reported in Table IX is based on 200 iterations of training, so a smaller training time can be achieved by moderately reducing the training iterations.

IV. Conclusion

In this article, we introduced the language model BERT-based hyperspectral image classification framework. The proposed HSI-BERT is primarily built on an MHSA mechanism that captures the global dependence of an arbitrary input region. With global dependencies learned by the MHSA approach, the proposed HSI-BERT not only captures global context but also attends to important locations with fewer redundancies (depends less on other locations). A richer global context leads to more informative global features, while the property of less redundancy enables HSI-BERT to only attend to fewer locations at one time (these locations have fewer dependencies with other locations), which eventually reduces the model complexity and leads to a faster and more accurate model convergence. The experimental comparisons show that the proposed HSI-BERT outperforms the state-of-the-art CNN methods in terms of both classification accuracy and computational time.

In addition to the competitive accuracy and the fast convergence speed, HSI-BERT enables the flexible and dynamic

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choice of input regions. Indeed, regions with more discriminative power can be explored in the future through our framework. Another important characteristic of our proposed method is its generalization ability. In this article, we investigated the generalization ability across input regions with different shapes, and in future work, we want to investigate the generalization ability across different HSI data.

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